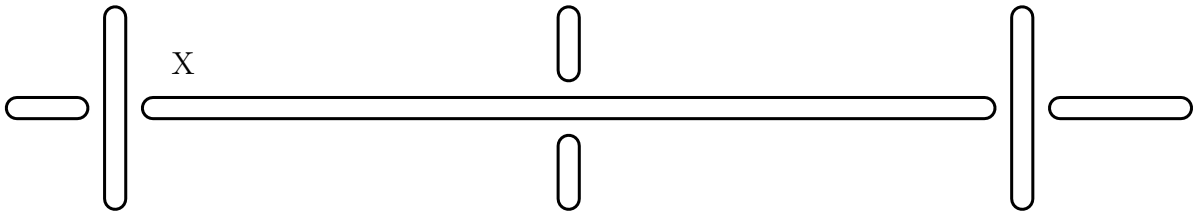
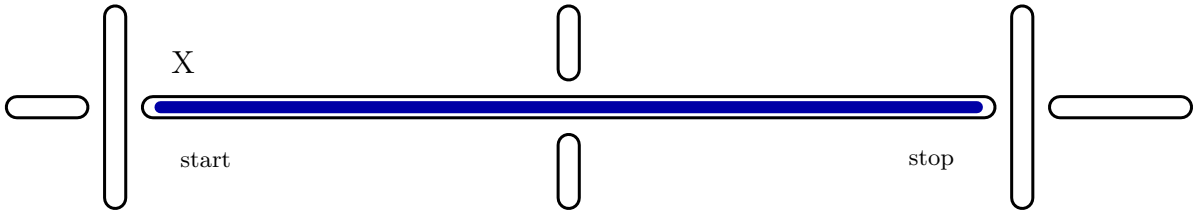
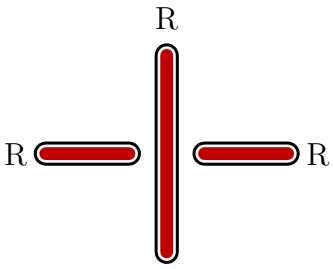
Keep cords on the table. Move them without cutting or passing strands through each other. Use red (R), blue (B), and green (G).



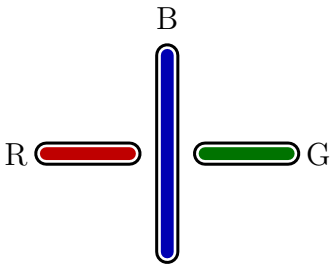
Follow X across the overpass to the next gap.



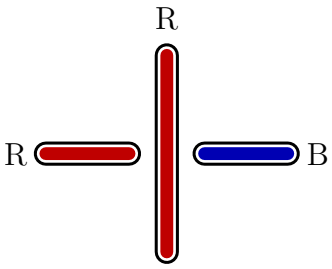
An arc runs between underpasses and continues through overpasses. A closed strand with no underpasses is one whole arc. Give each whole arc one color. At each crossing the colors must be all the same or all different.



valid

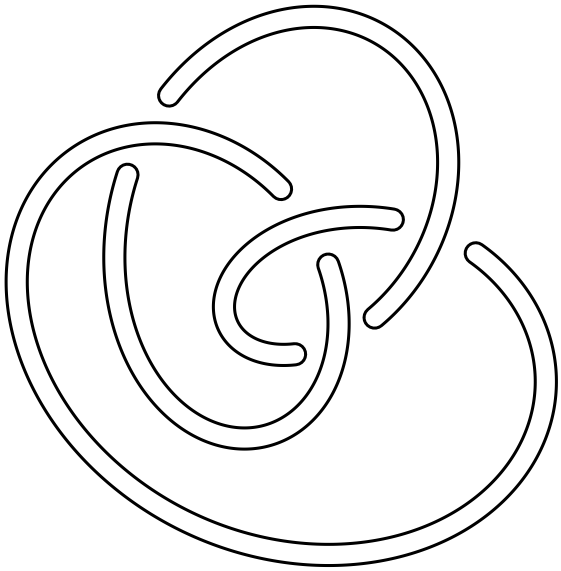


valid



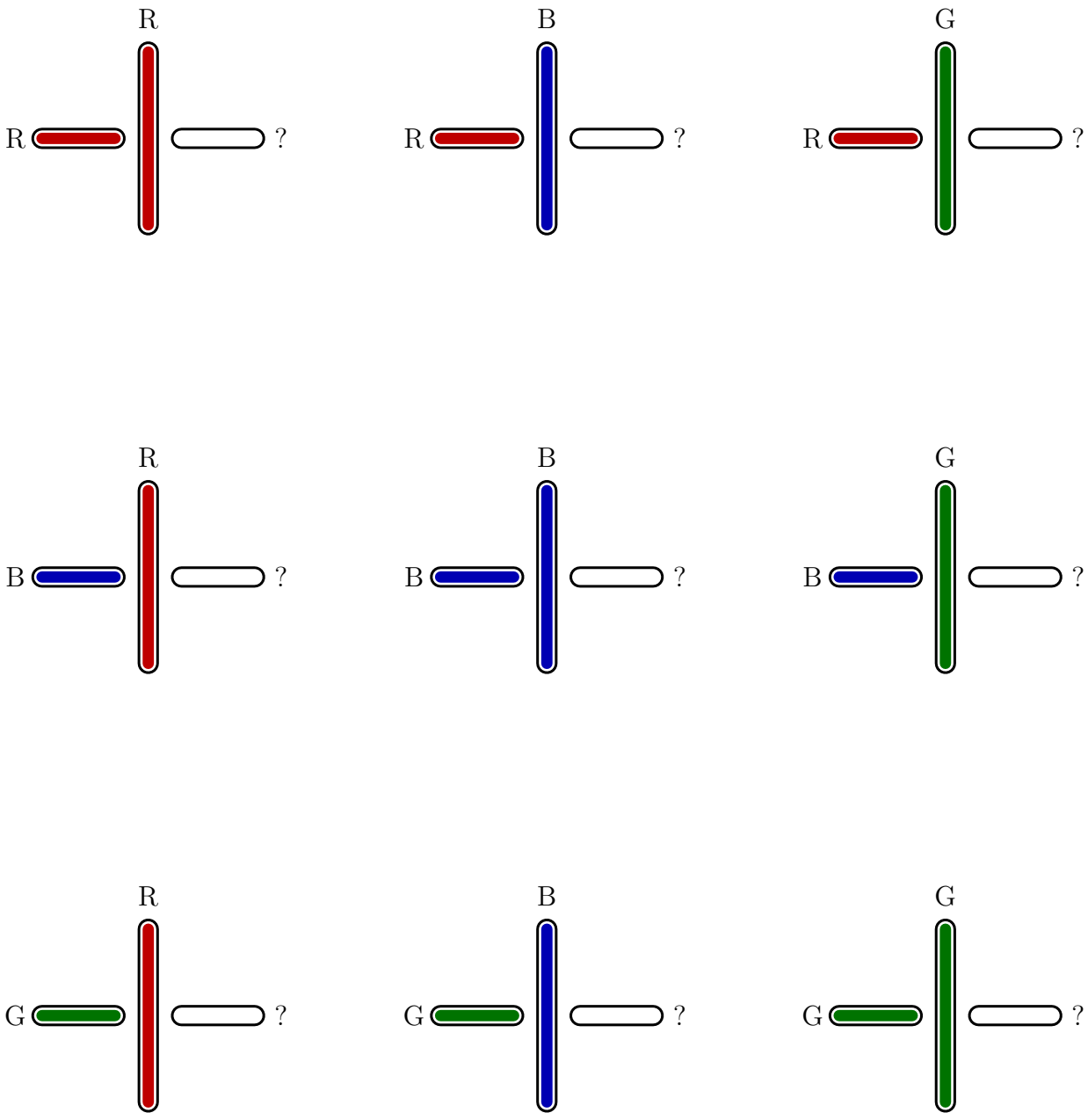
invalid

Problem 1: How many whole arcs does this loop have?

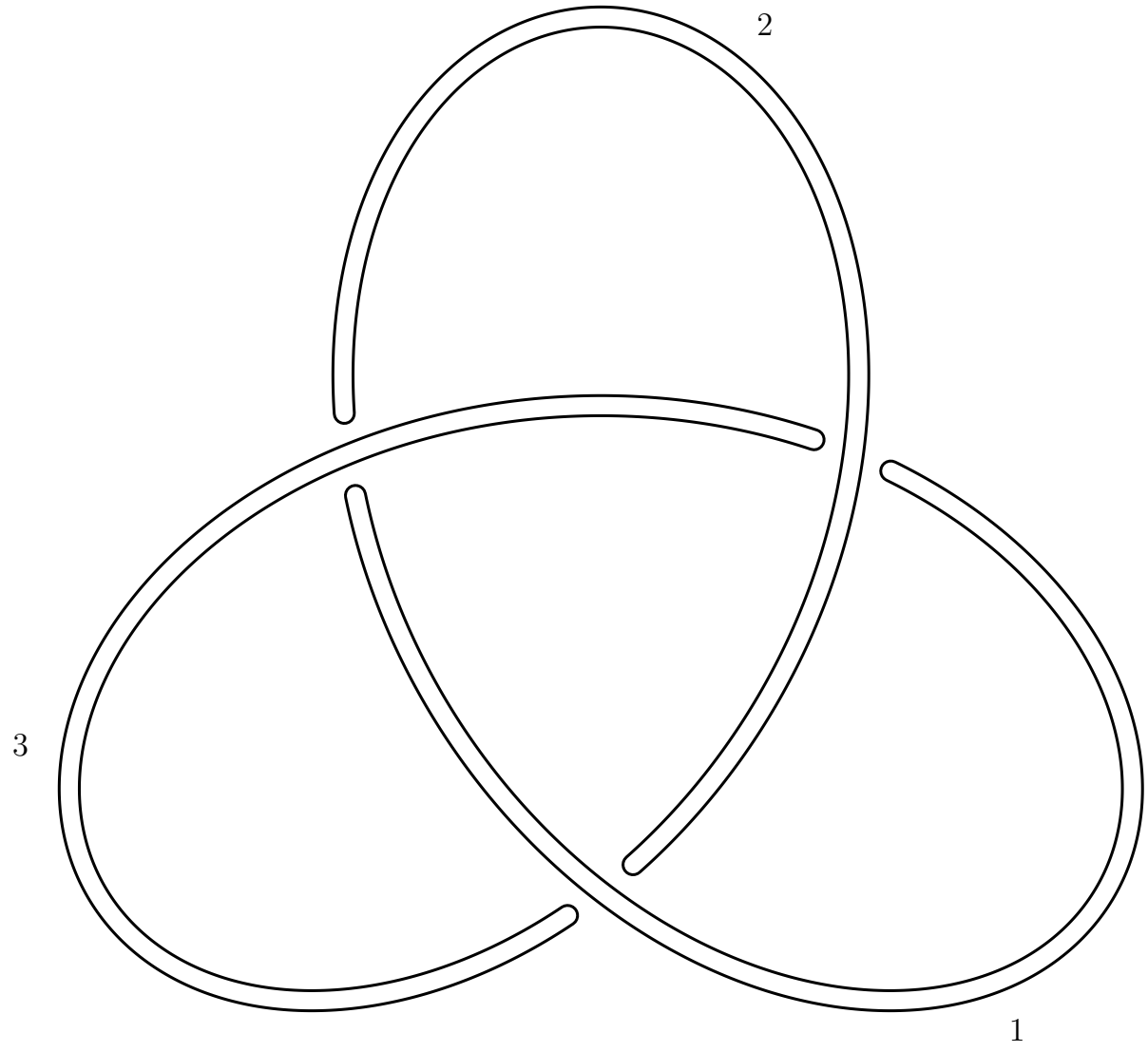


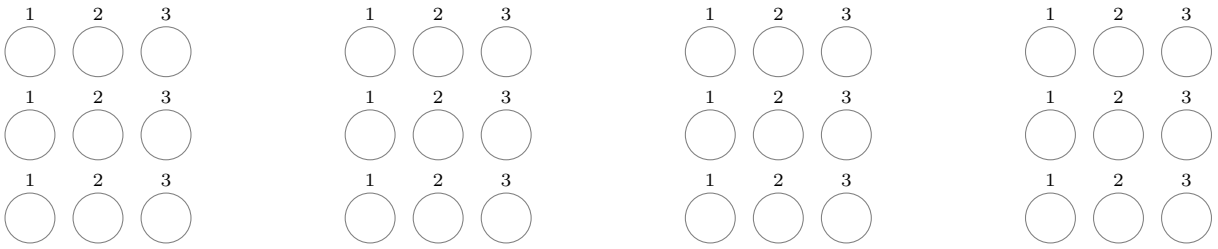
arcs: _____

Problem 2: Choose the missing colors so every crossing follows the rule. Can any question mark have two different answers?

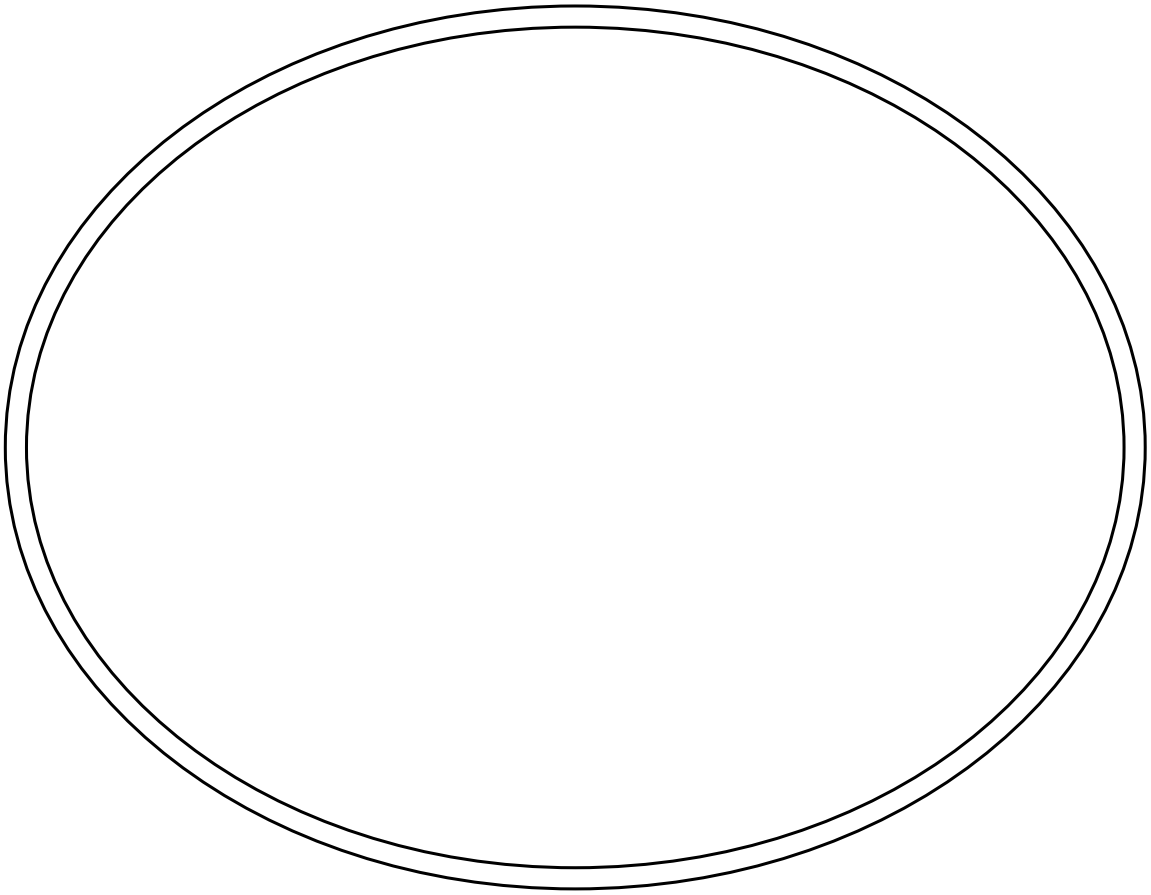


Problem 3: Find every valid coloring of this loop, using tracing paper for new tries. Record each coloring with color dots for whole arcs 1, 2, and 3.





Problem 4: Can you color this loop using more than one color while keeping one color on each whole arc?

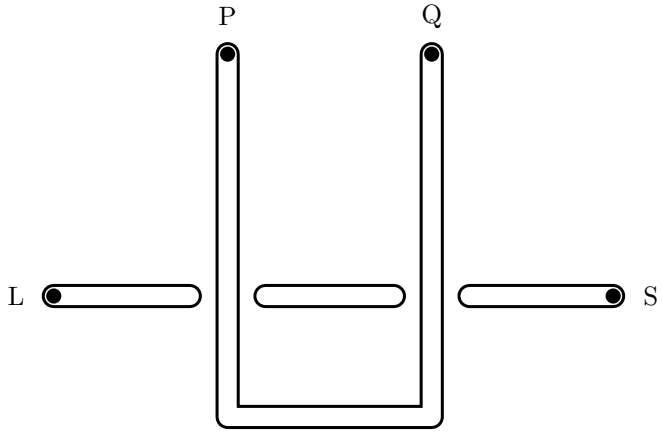


Problem 5: Try each of the four closed cords A, B, C and D in the tray. Can you move it into this plain loop shape without cutting or passing strands through one another?

A yes no B yes no C yes no D yes no

Problem 6: Keep each end dot its shown color. Which color choices let you color every arc of both strands and obey both crossings?

A strand may stop at an end dot in this picture. Use tracing paper for each try.



	L	P	S	Q
1	R	B	R	B
2	R	R	B	R
3	G	B	G	B
4	B	G	B	R
5	B	B	B	B
6	R	G	R	G